UST-Net: Unified Spatial–Spectral–Temporal Modeling for Adaptive Hyperspectral Object Tracking

Yanni Dong, *Senior Member, IEEE*, Jiaqi Zhang, Xiaole Han, and Jiawei Zhou

***Abstract*—Hyperspectral (HS) imagery provides material-sensitive spectral cues that complement spatial appearance and improve target discrimination when red–green–blue (RGB) observations are ambiguous. Existing HS trackers have explored spectral prompting and temporal modeling, yet spectrum-preserving prompt construction, recurrent target-state propagation, and hierarchical feature adaptation remain insufficiently coordinated. We propose UST-Net, a unified spatial–spectral–temporal framework that connects these processes within a shared transformer tracker. Its cross-modal prompt (CMP) module organizes co-registered false-color (FAC) and grouped multiband representations derived from the same HS cube into fixed-format prompts, preserving dependencies across spectral groups while accommodating different spectral configurations. The temporal state integration (TSI) module couples cross-frame Mamba hidden states with cross-stage semantic context and exchanges them with the prompt-enhanced token stream through paired injection and extraction operations. Conditioned on the resulting temporally refined tokens, the adaptive prompt injection (API) module generates stage-conditioned prompt context and applies it as residual modulation to downstream feature extraction. Experiments on the HOTC and HOT2024 datasets demonstrate that UST-Net achieves state-of-the-art performance across multiple spectral settings, with particularly strong results under occlusion and out-of-view (OV) conditions.**

***Index Terms*—Adaptive attention, cross-modal prompting, hyperspectral (HS) object tracking, memory-guided tracking, spatial–spectral–temporal modeling.**

I. Introduction

OBJECT tracking, the task of locating and following specific targets across video sequences, represents a fundamental challenge in computer vision with widespread applications in autonomous driving, surveillance, robotics, and human–computer interaction. While significant progress has been achieved in red–green–blue (RGB) tracking through correlation filters (CFs) [1], [2], deep learning [3], [4], and transformer-based approaches [5], [6], conventional RGB tracking faces inherent limitations in complex scenarios where targets exhibit similar visual appearances, camouflage effects, or undergo severe illumination variations (IVs).

Hyperspectral (HS) object tracking has emerged as a promising solution to address these challenges by leveraging rich spectral signatures across dozens or hundreds of wavelength bands instead of the three-channel RGB information. This capability proves particularly valuable when targets appear visually similar but have different material compositions [7], as each material exhibits distinct spectral characteristics. While RGB tracking relies on visual cues, HS imaging captures intrinsic material properties across the electromagnetic spectrum, providing enhanced discriminative power for target identification in complex environments. Applications span remote sensing, environmental monitoring, precision agriculture, defense surveillance, and industrial inspection [8], [9], [10], [11].

The evolution of HS object tracking has progressed from early CF-based methods such as material-based object tracking in HS videos (MHT) [7] and tracking in aerial HS videos using deep kernelized CFs (HKCFs) [12] to deep learning-based approaches [13], [14] and more recently to transformer-driven architectures [15], [16], [17]. Overall, these advances can be broadly categorized into three technical paradigms.

1) *Band selection and grouping methods*, including a band attention-aware ensemble network for HS tracking (BAE-Net) [13], band attention grouping-based Siamese object tracking network for HS videos (SiamBAG) [18], and exploring sparse mixture-of-experts for HS object tracking (HotMoE) [19], focus on reducing spectral dimensionality by selecting the most discriminative bands or grouping adjacent bands into RGB-like representations, thereby enabling the reuse of pretrained models. While effective in lowering computational costs, these methods often lose fine-grained spectral cues.
2) *Multimodal fusion approaches*, such as material-guided multiview fusion network for HS object tracking (MMF-Net) [14], HS video object tracker via fusing material and motion cues (SENSE) [20], and a transformer-based three-branch Siamese network for HS object tracking (TFTN) [21], seek to complement HS

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data by integrating it with other modalities (e.g., RGB or thermal) through various feature fusion strategies. These approaches enrich the representation space but depend heavily on the availability and alignment of additional modalities.

3) *Prompt learning methods*, represented by prompting for HS video tracking (PHTrack) [15], visual prompt for HS object tracking (VP-HOT) [22], and HS object tracking with dual-stream prompt (HDSP) [23], introduce prompt mechanisms to guide attention and enhance feature discrimination, showing promising adaptability across tasks.

Despite this progress, coordinating spatial–spectral representation, temporal state propagation, and hierarchical feature adaptation within a single HS tracker remains challenging for three interrelated reasons. Specifically,

1) *Configuration-Dependent Spectral Organization:* Existing band-selection, grouping, fusion, and prompt-based pipelines frequently compress high-dimensional spectra into a limited number of channels or specialize their encoders for predefined spectral ranges. A common representational interface is therefore required to organize co-registered false-color (FAC) and multiband observations while retaining fine-grained interband dependencies across spectral configurations.
2) *Limited Hierarchical State Propagation:* Many HS trackers remain centered on frame-local feature extraction or short-term historical aggregation. Without recurrent state propagation across multiple backbone depths, target representations can become unreliable under prolonged occlusion, out-of-view (OV) events, and subsequent reappearance.
3) *Temporally Decoupled Prompt Adaptation:* Prompts in existing HS trackers are commonly static or derived primarily from current-frame observations. Consequently, accumulated temporal context exerts limited influence on the adaptation of deeper feature representations as the target appearance and surrounding scene evolve. Collectively, these challenges call for a formulation that carries spectrum-preserving evidence into recurrent state evolution and subsequently reuses the accumulated context during hierarchical feature extraction.

To this end, we propose UST-Net, a unified spatial–spectral–temporal framework for HS object tracking. Cross-modal prompt (CMP) uses FAC tokens to query grouped multiband features and aggregates the aligned responses into fixed-format prompts. Four stage-aligned temporal state integration (TSI) modules then couple cross-frame Mamba states with cross-stage semantic context through paired injection and extraction, while adaptive prompt injection (API) converts the refined features into stage-conditioned residual modulation. This progressive pathway connects the current HS observation, accumulated target history, and multilevel visual representations within a shared transformer backbone. Experiments on two benchmarks demonstrate robust performance across multiple spectral settings.

The main contributions of this work are listed as follows.

1) We present UST-Net, a unified framework that organizes spectral prompting, recurrent state propagation, and stage-wise feature adaptation as successive stages of target-evidence refinement within a shared transformer tracker. This formulation provides a common downstream architecture for heterogeneous spectral configurations while allowing spectral cues and target history to shape representation learning at multiple backbone depths.
2) We design a CMP module that uses FAC tokens to query grouped multiband features derived from the same HS cube. Through band-wise cross-attention, 3-D spatial–spectral transformation, and quasi-recurrent spectral aggregation, CMP preserves interband dependencies and maps different spectral configurations into fixed-format prompts compatible with the shared backbone.
3) We develop a coupled temporal interaction and adaptive modulation mechanism comprising the TSI and API modules. TSI propagates Mamba hidden states across frames and semantic context across backbone stages through paired injection and extraction operations. API subsequently transforms the temporally refined features into stage-conditioned prompt context for residual modulation, allowing accumulated target evidence to influence feature extraction at successive depths.

II. Related Work

A. RGB Object Tracking

Over the past decade, RGB object tracking has progressed through several representative paradigms.

1) *CF-Based Methods:* CSK [1] exploits the circulant structure of training samples, KCF [24] extends correlation filtering through kernelized representations, and BACF [25] incorporates background-aware samples to improve target discrimination. Their Fourier-domain formulations enable efficient online tracking, but their reliance on relatively shallow appearance representations limits robustness under substantial appearance variation and occlusion.
2) *Deep Siamese Trackers:* SiamRPN [3], SiamRPN++ [26], and SiamFC++ [27] formulate tracking as offline-learned similarity matching and target localization, achieving a favorable balance between accuracy and efficiency. Their frame-pair-based inference, however, provides limited support for maintaining target states under drastic appearance changes, severe IV, and prolonged occlusion.
3) *Transformer-Based Trackers:* TransT [28], STARK [5], and OSTrack [4] employ attention mechanisms to strengthen template–search interaction and global dependency modeling. In particular, the one-stream formulation of OSTrack jointly performs feature extraction and relation modeling within a shared backbone, establishing a strong foundation for subsequent multimodal adaptation.

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Recent research has further extended transformer tracking toward foundation-model transfer and persistent temporal-context modeling. OneTracker [29] adapts a pretrained foundation tracker to RGB+X tasks through parameter-efficient prompting. ODTrack [30] establishes frame-to-frame associations by propagating compact temporal tokens online, while MambaLCT [31] recurrently aggregates long-term target-variation cues through state-space modeling.

Despite these advances, RGB-only trackers remain constrained by three-channel color observations, whereas existing RGB+X frameworks are generally tailored to predefined auxiliary modalities. These input formulations do not directly represent the densely sampled interband dependencies of a HS cube, leaving visually similar objects with different material compositions difficult to distinguish. This limitation motivates HS object tracking, which exploits discriminative spectral signatures distributed across dozens or hundreds of wavelength bands [32], [33].

B. HS Object Tracking

Research on HS object tracking has evolved along three complementary lines, reflecting a gradual transition from spectral compression toward increasingly flexible representation transfer.

1) *Band Selection and Grouping Methods:* These approaches reduce spectral redundancy to facilitate the efficient reuse of RGB-pretrained models. BAE-Net [13] estimates the discriminative importance of individual bands and assembles them into three-channel representations for pretrained backbones. SiamBAG [18] organizes spectral bands into attention-weighted groups, whereas SEE-NET [34] employs band-aware deep ensembles to aggregate complementary spectral cues. More recent designs further improve the flexibility of band utilization: HotMoE [19] performs adaptive expert selection through sparse routing, and SiamCAT [35] models spectral-channel variations using channel-adaptive feature aggregation within a dual-Siamese architecture. As the spectral axis is condensed, however, subtle interband relationships and material-specific signatures may be weakened. This tradeoff between computational efficiency and spectral fidelity has encouraged the use of multiple complementary representations.
2) *Multimodal Fusion Approaches:* Following this direction, multimodal trackers enrich target representations by combining HS observations with RGB, NIR, or other auxiliary views. MMF-Net [14] introduces a material-guided multiview fusion strategy that exploits material consistency across complementary representations. SENSE [20] associates HS material cues with RGB-derived motion information, while TFTN [21] adopts a three-branch Siamese architecture equipped with multihead attention to capture cross-modal dependencies. The additional representational capacity is accompanied by input-specific encoders and fusion topologies that are closely tied to prescribed modality pairs. Prompt learning has consequently emerged as a more flexible interface for transferring pretrained visual models across sensing configurations.
3) *Prompt Learning Methods:* Prompt-based adaptation incorporates modality-specific information through a compact set of trainable components while retaining the representational capacity of a pretrained tracker. In general multimodal tracking, ViPT [36] encodes auxiliary-modality cues as visual prompts and adapts an RGB-pretrained tracker with limited parameter updates. Within HS tracking, PHTrack [15] develops a hierarchical prompting scheme across multiple feature scales, VP-HOT [22] inserts learnable prompt tokens into successive feature-extraction stages, and HDSP [23] constructs dual-stream prompts for template and search representations. Related studies [37], [38] demonstrate the utility of prompt-based transfer for HS target representation. Extending this paradigm to heterogeneous HS observations requires prompt construction to preserve interband continuity, map different spectral configurations into a common token space, and evolve with the target state throughout a video sequence. Static learned tokens and frame-local cues provide only part of this functionality, bringing the coordination of spectral evidence and temporal context to the forefront.

C. Temporal Modeling for Visual Tracking

Temporal reasoning becomes particularly important once target representations are expected to evolve with appearance changes and scene dynamics. Early HS trackers primarily relied on template-update strategies, which conveyed recent appearance information without explicitly maintaining a long-term target state. Subsequent deep models introduced more structured temporal cues. The spectral–spatial–temporal attention network for HS tracking (SSTtrack) [39], for instance, jointly models spectral, spatial, and temporal conditions to promote temporal consistency. Transformer-based trackers further incorporate historical information through feature concatenation, short-term aggregation, or self-attention over temporal tokens. As the temporal window expands, the quadratic complexity of self-attention constrains the amount of historical context that can be retained and processed efficiently.

Selective state-space models reshape this computational tradeoff by compressing sequential history into recurrent latent states. Mamba [40] employs input-dependent state transitions to capture long-range dependencies with linear sequence complexity, and its visual extensions, including VMamba [41] and Vision Mamba [42], adapt this principle to visual representation learning. MCITrack [43] subsequently introduces Mamba hidden states as recurrent carriers of video-level context. Its contextual information fusion module exchanges historical memory with current visual features across multiple backbone blocks, illustrating hierarchical interaction between recurrent states and feature extraction.

State-space modeling has also entered HS tracking, where temporal evolution is intertwined with high-dimensional spectral variation. STTrack [44] develops a prompt Mamba module guided by spectral–temporal tokens. HyMamba [45] advances

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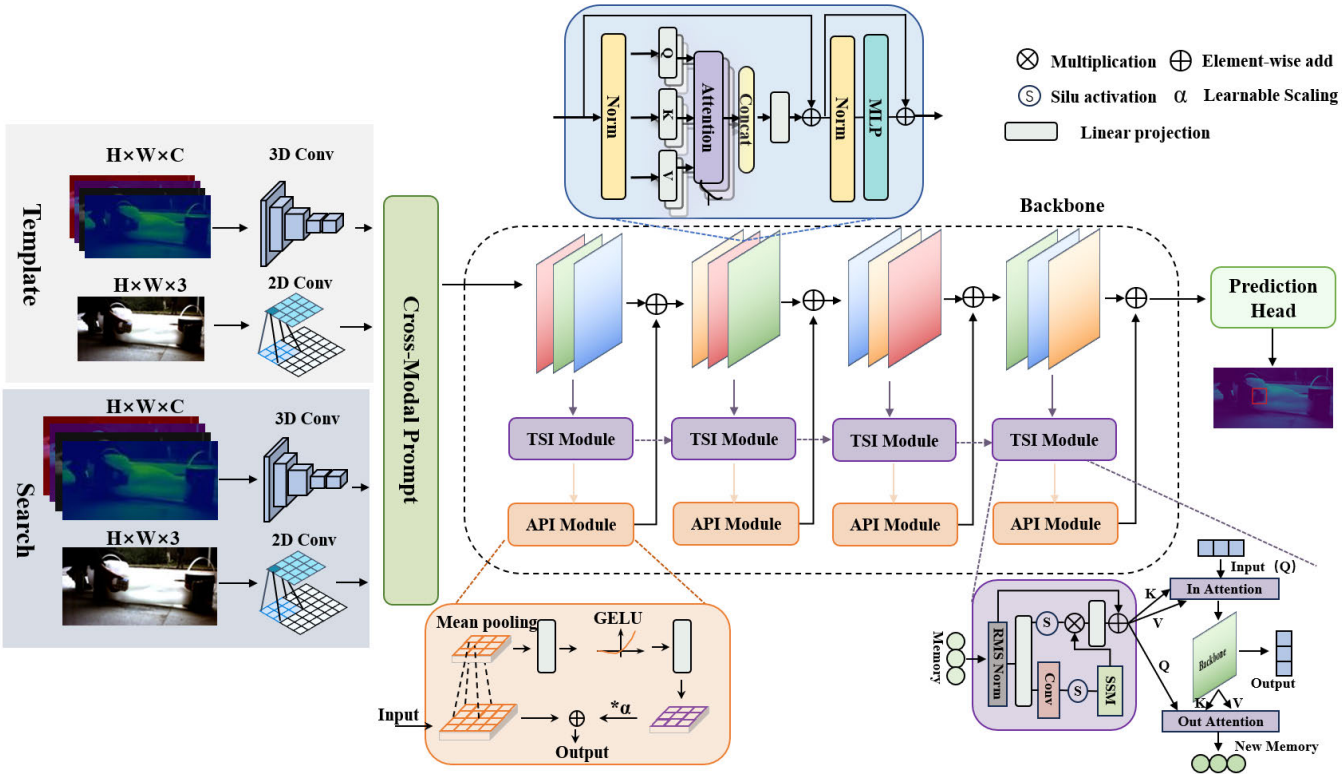


Fig. 1. Overall architecture and information flow diagram of UST-Net.

this line through spectral state integration and HS Mamba modules, using multidirectional state-space operations to coordinate spectral, cross-depth, and temporal dependencies. These approaches emphasize different information carriers: MCITrack propagates video-level visual context, STTrack employs spectral–temporal prompt tokens, and HyMamba maintains spectral and cross-depth states alongside temporal information.

Viewed jointly, these studies establish prompt-based transfer and state-space modeling as complementary approaches to spectral representation and temporal propagation. However, their coordination across backbone depths remains underexplored: existing methods do not jointly couple interband-aware prompts with recurrent target states and use the resulting context for hierarchical feature adaptation. This gap motivates UST-Net.

III. Method

This section presents the architecture and operation of UST-Net. We first outline the overall information flow and then describe the CMP, TSI, and API modules, followed by the prediction head and training objective.

A. Overall Framework

The overall architecture of UST-Net is illustrated in Fig. 1. The framework is built upon a shared transformer backbone comprising 12 encoder blocks partitioned into four consecutive stages. A CMP module provides the backbone with spatial–spectral prompts derived from the HS observation, while each stage is coupled with a TSI module and a corresponding API module for temporal-state interaction and stage-wise feature adaptation, respectively. The output of the final stage is decoded by a center-based prediction head.

UST-Net receives a template crop $\mathbf{z}$ of size 128×128 and a current-frame search crop $\mathbf{s}$ of size 256×256. Each crop is represented by a HS cube $\mathbf{H} \in \mathbb{R}^{H_{\text{img}} \times W_{\text{img}} \times C}$ and a co-registered FAC view $\mathbf{R} \in \mathbb{R}^{H_{\text{img}} \times W_{\text{img}} \times 3}$ synthesized from the same cube, where H_{img} and W_{img} denote the spatial dimensions of the input crop, and C is the number of spectral bands. The HS and FAC branches employ 3-D and 2-D patch embeddings, respectively, producing grouped spatial–spectral tokens $\mathbf{H}_{\text{token}} \in \mathbb{R}^{K \times N \times D}$ and spatial tokens $\mathbf{R}_{\text{token}} \in \mathbb{R}^{N \times D}$. Here, K denotes the number of spectral groups, N is the number of spatial patches in the corresponding crop, and D is the embedding dimension.

CMP organizes the two token sequences into a fixed-format prompt representation and incorporates it into the template–search token stream. The prompt-enhanced tokens are subsequently processed by the four backbone stages. At time t and stage i, TSI integrates the current tokens with the hidden memory M_i^{t-1} retained by the same stage from the preceding frame and the semantic context C_{i-1} propagated from the preceding stage within the current frame. The resulting hidden state M_i^t is preserved for the next frame, whereas the updated semantic context C_i advances to the next backbone stage. API derives stage-conditioned residual modulation from the temporally refined tokens and applies it to the feature stream forwarded downstream. After processing through all four stages, the final search region tokens are passed to the prediction head for target classification and localization.

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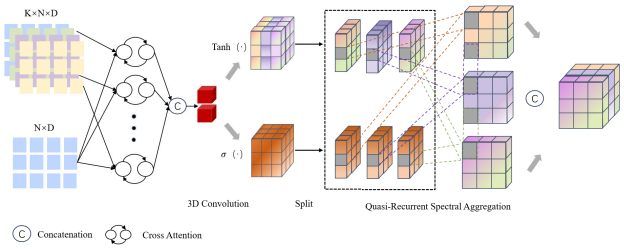


Fig. 2. Architecture of the CMP module. The module processes HS and FAC tokens through band-wise cross-attention, 3-D spatial-spectral transformation, and quasi-recurrent spectral aggregation to generate configuration-consistent spatial–spectral prompt embeddings.

B. CMP Module

CMP converts co-registered multiband and FAC representations from the same HS cube into a spatial–spectral prompt compatible with the shared backbone. The FAC features provide a common spatial reference for querying grouped multiband features. CMP preserves group-specific responses during cross-modal interaction and aggregates the spectral-group axis only after intergroup dependencies have been modeled.

As illustrated in Fig. 2, CMP receives two token sequences produced by the 2-D and 3-D patch embedding modules. Let $\mathbf{H}_{\text{token}} \in \mathbb{R}^{K\times N\times D}$ and $\mathbf{R}_{\text{token}} \in \mathbb{R}^{N\times D}$ denote the tokenized representations of the HS and FAC views, respectively, where K is the number of spectral groups, N denotes the number of spatial patches, and D is the embedding dimension.

CMP first performs band-wise cross-attention between each spectral group $\mathbf{H}^k_{\text{token}} \in \mathbb{R}^{N\times D}(1 \le k \le K)$ and the FAC sequence $\mathbf{R}_{\text{token}}$. The FAC tokens serve as queries, while the tokens from the kth HS group provide the keys and values. The attention computation is formulated as

$$\mathbf{F}^k = \text{Softmax}\left(\frac{\left(\mathbf{R}_{\text{token}}\mathbf{W}_Q\right)\left(\mathbf{H}^k_{\text{token}}\mathbf{W}_K\right)^\top}{\sqrt{d}}\right)\cdot\left(\mathbf{H}^k_{\text{token}}\mathbf{W}_V\right)\mathbf{W}_O \tag{1}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D\times d}$ are linear projections and $\mathbf{W}_O \in \mathbb{R}^{d\times D}$ restores the feature dimension. By applying the same FAC query sequence to all spectral groups, the resulting features maintain a consistent spatial organization while retaining complementary spectral responses.

The attended outputs from all spectral groups are concatenated along the group axis

$$\mathbf{F}_{\text{spatial}} = \text{Concat}\left(\mathbf{F}^1, \mathbf{F}^2, \ldots, \mathbf{F}^K\right) \in \mathbb{R}^{K\times N\times D}. \tag{2}$$

To capture local spatial–spectral dependencies, we reshape this tensor into a 3-D feature volume $\hat{\mathbf{F}} \in \mathbb{R}^{D\times K\times H\times W}$, where $H \times W = N$ denotes the spatial patch grid. This volumetric arrangement exposes both the ordered spectral-group dimension and the 2-D spatial neighborhood to subsequent 3-D processing. Two depth-wise 3-D convolutions are then applied to extract local spatial–spectral features while maintaining computational efficiency

$$\mathbf{Z} = \tanh\left(\mathbf{W}_z * \hat{\mathbf{F}}\right), \quad \mathbf{Y} = \sigma\left(\mathbf{W}_y * \hat{\mathbf{F}}\right) \tag{3}$$

where $*$ denotes 3-D convolution, and $\mathbf{Z}, \mathbf{Y} \in \mathbb{R}^{D\times K\times H\times W}$ represent the candidate and gate features, respectively. The tanh activation produces bounded candidate responses, while the sigmoid activation generates gate values in $[0, 1]$ for the subsequent propagation of spectral information.

The 3-D convolutions capture interactions within local spatial–spectral neighborhoods. To further propagate information across the ordered spectral groups, we apply a quasi-recurrent aggregation mechanism [46] along the spectral dimension. For each spatial location (i, j) and feature channel d, the aggregation proceeds as

$$\mathbf{h}_0 = \mathbf{0}, \quad \mathbf{h}_k = \mathbf{y}_k \odot \mathbf{h}_{k-1} + (1 - \mathbf{y}_k) \odot \mathbf{z}_k, \quad k = 1, 2, \ldots, K \tag{4}$$

where $\mathbf{y}_k, \mathbf{z}_k, \mathbf{h}_k \in \mathbb{R}^{D\times H\times W}$ represent the gate features, candidate features, and hidden states at the kth spectral group, respectively. Here, $\odot$ denotes element-wise multiplication. At each step, the gate balances the information retained from the preceding groups and the candidate response extracted from the current group. The recurrent state therefore accumulates intergroup spectral evidence while preserving its spatial correspondence.

The final prompt embedding is obtained as

$$\mathbf{P} = \mathbf{h}_K \in \mathbb{R}^{D\times H\times W}. \tag{5}$$

This prompt embedding is then reshaped to $\mathbb{R}^{N\times D}$ to match the transformer token format and injected into the token stream before each transformer stage. In the first stage, the full spatial–spectral prompt is used, while in deeper stages, only spatial alignment is performed to reduce computational overhead.

Although VIS, NIR, and RedNIR inputs contain different numbers of bands and spectral groups, aggregating the group axis produces a configuration-consistent prompt of shape $\mathbb{R}^{N\times D}$, allowing the shared backbone, TSI, and API modules to operate without architectural reconfiguration.

C. TSI Module

The TSI module integrates recurrent target history with hierarchical backbone processing through a stage-aligned injection–extraction mechanism. As illustrated in Fig. 3, TSI maintains two complementary state carriers with distinct propagation routes: 1) the hidden memory $\mathbf{M}_i^{t-1}$, which is retained by the ith stage and propagated across consecutive frames; and 2) the semantic context $\mathbf{C}_{i-1}$, which conveys the current-frame representation from the preceding backbone stage. The former provides a persistent carrier of temporal information, while the latter transfers progressively enriched semantics across backbone depth.

To coordinate these two propagation axes, the 12-layer backbone is divided into four semantic stages comprising layers 0–2, 3–5, 6–8, and 9–11, respectively. Each stage is paired with a corresponding TSI module. At stage i and frame t, the recurrent state follows the frame-wise path $\mathbf{M}_i^{t-1} \to \mathbf{M}_i^t$ at a fixed backbone depth, whereas the semantic context follows the stage-wise path $\mathbf{C}_{i-1} \to \mathbf{C}_i$ within the current frame. This stage-aligned organization allows historical target information

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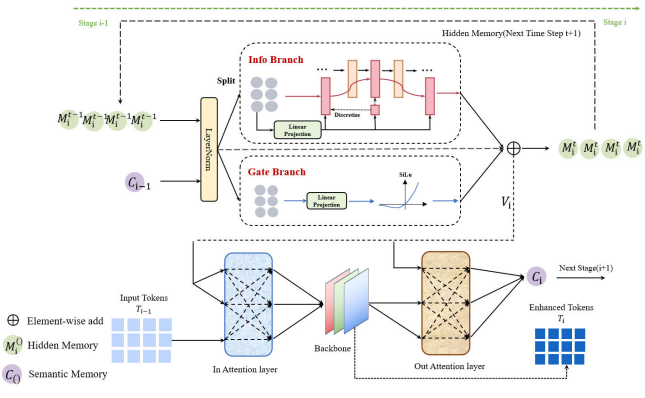


Fig. 3. Overview of the TSI module. At stage i and time step t, the module receives semantic context C_{i-1} and hidden memory M_i^{t-1}. The Mamba block updates the hidden memory to M_i^t and produces memory features V_i. These features are injected into the token stream through in-attention, followed by backbone processing to obtain enhanced tokens. The tokens then interact with V_i throughout-attention to produce the updated semantic context C_i, forming a hierarchical injection–update loop across stages and frames.

and current-frame semantics to interact repeatedly as the token representation progresses through the backbone.

Each TSI module combines a Mamba residual block with paired in-attention and out-attention operators. The temporal refinement at stage i is formulated as

$$\mathbf{V}_i, \mathbf{M}_i^t = \text{Mamba}_i\left(\mathbf{C}_{i-1}, \mathbf{M}_i^{t-1}\right) \tag{6}$$

$$\mathbf{T}_i' = \text{Att}_i^{\text{in}}\left(\mathbf{T}_{i-1}, \mathbf{V}_i, \mathbf{V}_i\right) \tag{7}$$

$$\mathbf{T}_i = \text{Block}_i\left(\mathbf{T}_i'\right) \tag{8}$$

$$\mathbf{C}_i' = \text{Att}_i^{\text{out}}\left(\mathbf{V}_i, \mathbf{T}_i, \mathbf{T}_i\right) + \mathbf{V}_i \tag{9}$$

$$\mathbf{C}_i = \mathbf{C}_i' + \text{FFN}_i\left(\mathbf{C}_i'\right). \tag{10}$$

Here, $\mathbf{C}_{i-1}$ represents the semantic context received from the preceding stage within frame t, while $\mathbf{M}_i^{t-1}$ is the hidden state retained by the ith TSI module from frame $t-1$. The Mamba block combines these inputs to produce the memory feature $\mathbf{V}_i$ and the updated cross-frame state $\mathbf{M}_i^t$ (6). In-attention then injects $\mathbf{V}_i$ into the prompt-enhanced input tokens $\mathbf{T}_{i-1}$, yielding the temporally conditioned sequence $\mathbf{T}_i'$ (7). The corresponding backbone stage transforms $\mathbf{T}_i'$ into $\mathbf{T}_i$ (8). Out-attention subsequently uses $\mathbf{V}_i$ as the query and $\mathbf{T}_i$ as the keys and values, extracting the current-stage evidence into $\mathbf{C}_i'$. A residual FFN produces the semantic context $\mathbf{C}_i$ forwarded to stage $i+1$ [(9) and (10)]. Meanwhile, $\mathbf{M}_i^t$ remains associated with stage i and is retrieved when frame $t+1$ is processed. For cross-frame propagation, TSI retains the updated recurrent state rather than an explicit set of feature tokens from a predefined number of preceding frames. At each stage, $\mathbf{M}_i^t$ recursively summarizes the accumulated cross-frame information available to that stage and replaces $\mathbf{M}_i^{t-1}$ after the current frame has been processed. The structure of the temporal memory therefore remains fixed as the sequence proceeds, and TSI introduces no explicit temporal-window or memory-length hyperparameter.

Accordingly, $\mathbf{M}_i^t$ propagates across frames, $\mathbf{C}_i$ across backbone stages, and $\mathbf{T}_i$ to the corresponding API module. This stage-aligned routing allows accumulated target evidence to condition prompt-enhanced representations at multiple backbone depths.

1) Mamba Residual Block Architecture: The Mamba residual block provides the recurrent state update used by each TSI module. Given the paired state inputs $\mathbf{C}_{i-1}$ and $\mathbf{M}_i^{t-1}$, each represented in $\mathbb{R}^{B\times L\times D}$, where B is the batch size, L is the sequence length, and D is the feature dimension; the input is first rms-normalized and then split into two functional branches for parallel processing.

Information Branch. The first half of the normalized input is projected and passed through a depthwise 1-D convolution to extract local dependencies within the state sequence. Let $\mathbf{x}_t$ denote the normalized input feature supplied to the discretized state-space model. The information branch updates the recurrent state through the following SSM:

$$\mathbf{x}_t^{\text{conv}} = \text{SiLU}\left(\text{DWConv1D}\left(\mathbf{x}_t\right)\right) \tag{11}$$

$$\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{x}_t^{\text{conv}} \tag{12}$$

$$\mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{x}_t^{\text{conv}}. \tag{13}$$

Here, $\mathbf{h}_t$ is the updated SSM state and $\mathbf{y}_t$ is the corresponding state-conditioned output. At the TSI level, the state retained after processing the current input corresponds to the stage-wise temporal memory $\mathbf{M}_i^t$. The matrices $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ denote the discretized state-transition and input matrices, while $\mathbf{C}$ and $\mathbf{D}$ parameterize the state readout and residual input mapping, respectively. The SSM matrix $\mathbf{C}$ is distinct from the stage-wise semantic context $\mathbf{C}_i$ used in the TSI formulation.

Gating Branch. The second half of the normalized input is projected through a learnable linear transformation $\mathbf{W}_z \in \mathbb{R}^{D\times D}$, followed by a SiLU activation to generate a content-dependent gating signal

$$\mathbf{z}_t = \text{SiLU}\left(\mathbf{W}_z\mathbf{x}_t\right). \tag{14}$$

Output Composition. The information and gating branches are combined through element-wise modulation and a residual connection

$$\mathbf{o}_t = \mathbf{y}_t \odot \mathbf{z}_t + \mathbf{x}_t \tag{15}$$

where the residual term $\mathbf{x}_t$ maintains a direct information path through the block. The gated output $\mathbf{o}_t$ is linearly projected to produce the memory feature $\mathbf{V}_i$, while the updated internal state $\mathbf{h}_t$ is retained as the hidden memory $\mathbf{M}_i^t$, following the state transition in (12).

The dual-branch structure separates recurrent state evolution from content-dependent output modulation. Embedded at the four semantic stages, it supplies each TSI module with a persistent cross-frame state and a memory feature for interaction with the current token stream. Together with the in-attention and out-attention operators, this design coordinates temporal propagation through $\mathbf{M}_i^t$ and hierarchical semantic transfer through $\mathbf{C}_i$, supporting stable target representation under appearance changes, occlusion, and target reappearance.

D. API Module

API follows TSI at each backbone stage and converts the temporally refined representation into stage-conditioned residual modulation. Specifically, TSI first updates the hidden

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memory $\mathbf{M}_i^t$ and produces the refined tokens $\mathbf{T}_i$. API then takes $\mathbf{T}_i$ as its input, denoted by $\mathbf{X}$ in this section. Since $\mathbf{T}_i$ has already interacted with the cross-frame memory $\mathbf{M}_i^{t-1}$ and the cross-stage context $\mathbf{C}_{i-1}$, the modulation generated by API is conditioned jointly on the current feature distribution and the accumulated target history. Temporal-state recurrence remains within TSI, while API provides a feed-forward adaptation path for the downstream feature stream.

Each API instance comprises prompt generation and prompt injection. Prompt generation compresses the refined token sequence into a global descriptor and expands it into a set of latent prompt components. Prompt injection subsequently scales and aggregates these components into a stage-conditioned context vector, which is added to the outgoing token sequence through a residual connection. In this way, the temporally enriched representation produced at one stage can influence feature extraction at the next backbone depth.

1) Prompt Generation: Given the temporally refined token sequence $\mathbf{X} \in \mathbb{R}^{B\times N\times D}$, where B is the batch size, N is the number of spatial tokens, and D is the feature embedding dimension, API first summarizes its feature distribution through mean pooling

$$\mathbf{g} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{X}_i \in \mathbb{R}^{B\times D} \tag{16}$$

where the summation index i enumerates the token positions. The resulting descriptor $\mathbf{g}$ provides a compact representation of the current stage output, including the temporal information incorporated by TSI.

The descriptor is then processed by a lightweight feed-forward network to generate M latent prompt components

$$\mathbf{T} = \text{Reshape}\left(\text{GELU}\left(\text{LN}\left(\mathbf{g}\right)\mathbf{W}_1\right)\mathbf{W}_2\right) \in \mathbb{R}^{B\times M\times D} \tag{17}$$

where $\mathbf{W}_1 \in \mathbb{R}^{D\times D/2}$ and $\mathbf{W}_2 \in \mathbb{R}^{D/2\times(M\times D)}$ are learnable parameters, and LN denotes LayerNorm. Here, $\mathbf{T}$ is the generated prompt tensor, and M is the number of latent prompt components, distinct from the TSI hidden memory $\mathbf{M}_i^t$. We set $M = 8$ in all API instances, yielding $\mathbf{T} \in \mathbb{R}^{B\times 8\times D}$. Because these components are consolidated into $\mathbf{T}_{\text{context}}$ rather than concatenated with $\mathbf{X}$, M does not change the downstream token number N or attention sequence length.

2) Prompt Injection: The generated prompt tensor is first scaled by a learnable parameter α

$$\mathbf{T}_{\text{scaled}} = \alpha \cdot \mathbf{T} \tag{18}$$

where α is a learnable scalar initialized to 0.01. The small initial value keeps the newly introduced residual branch close to the original feature path at the beginning of optimization, while subsequent learning adjusts the modulation magnitude for each API instance.

The scaled prompt components are then aggregated into a global modulation vector

$$\mathbf{T}_{\text{context}} = \frac{1}{M}\sum_{j=1}^{M}\mathbf{T}_{\text{scaled},j} \in \mathbb{R}^{B\times D}. \tag{19}$$

This operation consolidates the M latent components into a stage-conditioned context with the same feature dimension as each input token. The context is broadcast along the sequence dimension and added to the input through residual modulation

$$\tilde{\mathbf{X}} = \mathbf{X} + \text{Broadcast}\left(\mathbf{T}_{\text{context}}, N\right) \tag{20}$$

where the broadcast operation expands $\mathbf{T}_{\text{context}} \in \mathbb{R}^{B\times D}$ along the sequence dimension to match $\mathbf{X} \in \mathbb{R}^{B\times N\times D}$. The residual formulation preserves the number and dimensionality of the tokens while conditioning every token on the stage-level context derived from the TSI-refined representation. For the first three backbone stages, $\tilde{\mathbf{X}}$ is forwarded to the subsequent stage; after the fourth stage, the modulated search-region tokens are delivered to the prediction head.

3) Remarks: API introduces no additional recurrent state; its prompts are regenerated from the current TSI-refined tokens at each stage. TSI carries recurrent states across frames and semantic context across stages, whereas API converts the resulting representation into residual guidance for downstream feature refinement.

E. Prediction Head and Loss Function

Following the final API module, the search-region tokens are reshaped into a 2-D feature map and passed to the center-based prediction head adopted from OSTrack [4]. The head comprises three convolutional branches that predict a center-score map $\mathbf{S} \in \mathbb{R}^{1\times H_s/P\times W_s/P}$, a bounding-box size map $\mathbf{B} \in \mathbb{R}^{2\times H_s/P\times W_s/P}$, and a center-offset map $\mathbf{O} \in \mathbb{R}^{2\times H_s/P\times W_s/P}$, respectively. Here, H_s and W_s denote the height and width of the search region, and P is the downsampling ratio. The target center is determined from the peak response in $\mathbf{S}$, and the corresponding predictions in $\mathbf{B}$ and $\mathbf{O}$ are used to recover the final bounding box.

The center-score branch is supervised using focal loss [47], while bounding-box regression is optimized with L_1 loss and GIoU loss [48]. Each frame contributes an individual tracking loss during video-level training. The total loss is

$$\mathcal{L} = \sum_{i=1}^{n}\left(\lambda_c\mathcal{L}_i^{\text{cls}} + \lambda_l\mathcal{L}_i^{l} + \lambda_g\mathcal{L}_i^{g}\right). \tag{21}$$

$\mathcal{L}_i^{\text{cls}}$, $\mathcal{L}_i^{l}$, and $\mathcal{L}_i^{g}$ denote the focal classification, L_1, and GIoU losses for frame i, respectively. We set $\lambda_c = 1$, $\lambda_l = 5$, and $\lambda_g = 2$.

IV. Experiments

A. Implementation Details

1) Model and Training: We adopt the ViT-based OSTrack as our foundation model, which is pretrained on large-scale color video datasets and then fine-tuned on FAC sequences from the HOT2024 benchmark, where sequences overlapping with the HOT2020 validation set are removed. During HS training, we follow the protocol in [4] and freeze all parameters of the backbone and prediction head. Only the newly introduced modules—the CMP module, the TSI module, and the API module—are trained. This results in 53.4 million trainable parameters, while the total number of model parameters is 145.9 million. The overall computational complexity of the proposed model is 29.1 GFLOPs for a single template–search

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TABLE I
EFFICIENCY COMPARISON IN TERMS OF COMPUTATIONAL COMPLEXITY (FLOPS)

Trackers	SEE-Net	OSTrack	SSTTrack	AQATrack	SiamRPN++
FLOPs (G)	244.2	29.0	315.0	25.8	48.9
Trackers	BAE-Net	SeqTrack	SiamBAN	SiamCAR	**UST-Net**
FLOPs (G)	314.9	148.0	73.6	69.2	**29.1**

pair. A comparison of computational complexity with representative trackers is provided in Table I.

The four stage-aligned TSI modules employ independently parameterized Mamba residual blocks, with one block assigned to each stage. Each block adopts RMSNorm prenormalization, an expansion ratio of 2, an SSM state dimension of 16, a depth-wise 1-D convolution with a kernel size of 3, and a time-step projection rank of 48.

Training is conducted on a single NVIDIA RTX 4090 GPU for 20 epochs. Each epoch contains 5600 template–search pairs, with a batch size of 16. The template and search regions are cropped to 128×128 and 256×256 pixels, respectively. We use the AdamW optimizer with an initial learning rate of 8×10^{-5} and a weight decay of 0.0001. The learning rate is decayed by a factor of 10 after 16 epochs.

2) Inference: Inference is conducted at 15 FPS using a single NVIDIA RTX 4090 GPU, with the input sequence processed frame by frame. A stage-indexed recurrent state cache, aligned with the four semantic stages of the backbone, is maintained throughout each video sequence. The cache is initialized as empty at the beginning of a new sequence. When frame t is processed, the ith TSI module retrieves the hidden state $\mathbf{M}_i^{t-1}$ associated with the same stage and integrates it with the current-frame representation. The updated state $\mathbf{M}_i^t$ is then written back to the corresponding cache entry and retrieved when frame $t+1$ is processed. Each update replaces the preceding state rather than appending a new set of frame features. Consequently, the cache maintains one current hidden state for each TSI stage, remains fixed in structure as the sequence length increases, and requires no predefined temporal window. This recurrent update allows historical context to be propagated across consecutive frames without maintaining an expanding collection of past-frame tokens.

B. Benchmark Datasets

HS Dataset: To comprehensively evaluate the proposed method, two benchmark datasets are employed in the experiments. The details of these datasets are described as follows.

1) HOTC Dataset: The HOTC dataset [7], released as part of the HS Object Tracking Challenge, is the first publicly available HS video dataset specifically designed for object tracking. It comprises 75 video sequences, with 40 sequences used for training and 35 for testing. The HS sequences were recorded at 25 FPS using a commercial snapshot-mosaic HS camera, while the corresponding RGB sequences were independently captured at the same frame rate from closely aligned viewpoints. Each sequence is provided in three modalities: HS, FAC, and RGB. The HS videos contain 16 spectral bands covering wavelengths from 460 to 600 nm, and the FAC videos are generated from the HS observations using a CIE-based spectral-to-RGB conversion method.

The sequences are annotated with 11 attributes that reflect various challenging CN scenarios: Background Clutter (BC), Motion Blur (MB), Deformation (DEF), Fast Motion (FM), In-Plane Rotation (IPR), Out-of-Plane Rotation (OPR), Scale Variation (SV), IV, Occlusion (OCC), OV, and Low Resolution (LR). These comprehensive annotations, along with the multimodal video formats, make HOTC a rigorous and diverse benchmark for evaluating the robustness and generalization capabilities of HS object tracking algorithms.

2) HOT2024 Dataset: The HOT2024 dataset, released as part of the HS Object Tracking Challenge 2024, consists of three subsets: HOT2024-VIS, HOT2024-NIR, and HOT2024-RedNIR. The videos were recorded at 25 FPS using three XIMEA snapshot-mosaic HS cameras dedicated to the VIS, NIR, and RedNIR spectral ranges. Each raw HS frame was acquired as a 2-D spectral mosaic and subsequently reconstructed into a 3-D HS cube after calibration. Each HS frame is accompanied by a FAC image generated using the CIE color-mapping method [13].

1) HOT2024-VIS consists of HS videos with 16 spectral bands, including 111 training sequences and 67 testing sequences.
2) HOT2024-NIR contains 25 spectral bands per frame, with 70 videos for training and 30 for testing.
3) HOT2024-RedNIR includes videos with 15 spectral bands, providing 36 training sequences and 20 testing sequences.

In total, the dataset comprises 217 training sequences and 117 testing sequences. Bounding box annotations are provided for each frame to support supervised learning.

C. Comparison With SOTA Trackers

1) Results on HOTC Dataset: To comprehensively evaluate the effectiveness of our proposed UST-Net, we conduct extensive experiments on the HOTC dataset. We compare our method with SOTA RGB-based and HS trackers. The quantitative results are summarized in Tables II and III. A detailed analysis is provided as follows.

a) RGB trackers comparison: Table II reports the performance of 22 representative RGB trackers, including CSK [1], adaptive color attributes for real-time visual tracking (CN) [2], a scale adaptive kernel CF tracker with feature integration (SAMF) [49], KCF [24], Staple [50], BACF [25], adaptive regularization for CF tracking (ARCF) [51], toward high-performance visual tracking for UAV with automatic spatio-temporal regularization (AutoTrack) [52], SiamRPN [3], SiamRPN++ [26], target-aware tracking with Siamese box adaptive network (SiamBAN) [53], SiamFC++ [27], graph attention tracking (SiamGAT) [54], STARK [5], Siamese fully convolutional classification and regression for visual tracking (SiamCAR) [55], generalized relation modeling for transformer tracking (GRM) [6], sequence to sequence learning for visual object tracking (SeqTrack) [56], exploring spatial-modular attention for visual tracking (SMAT) [57], autoregressive queries for adaptive tracking with spatio-temporal

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TABLE II

PARALLEL COMPARISON WITH SOTA RGB TRACKERS ON HOTC DATASET. THE BEST TWO RESULTS ARE SHOWN IN RED AND BLUE FONTS

NO.	Tracker	Venue	Features/Backbones/Tags	RGB Pre	RGB Suc	FAC/HS Pre	FAC/HS Suc
1	CSK [1]	ECCV 2012	I	0.574	0.330	0.612	0.342
2	CN [2]	CVPR 2014	CN+I	0.644	0.379	0.641	0.378
3	SAMF [49]	ECCV 2015	HOG+CN+I	0.691	0.418	0.657	0.387
4	KCF [24]	TPAMI 2015	HOG	0.611	0.376	0.584	0.357
5	Staple [50]	CVPR 2016	HOG+CN	0.800	0.518	0.768	0.507
6	BACF [25]	ICCV 2017	HOG	0.792	0.533	0.816	0.544
7	ARCF [51]	ICCV 2019	HOG+CN+I	0.817	0.565	0.796	0.542
8	AutoTrack [52]	CVPR 2020	HOG+CN+I	0.830	0.535	0.816	0.534
9	SiamRPN [3]	CVPR 2018	AlexNet/Default	0.900	0.593	0.756	0.486
10	SiamRPN++ [26]	CVPR 2019	ResNet-50/L-T	0.866	0.621	0.761	0.531
11	SiamBAN [53]	CVPR 2020	ResNet-50/OTB	0.851	0.610	0.861	0.587
12	SiamFC++ [27]	AAAI 2020	GoogLeNet/OTB	0.863	0.636	0.818	0.579
13	SiamGAT [54]	CVPR 2021	GoogLeNet/LaSOT	0.787	0.541	0.768	0.501
14	STARK [5]	ICCV 2021	ResNet-50	0.893	0.638	0.813	0.579
15	SiamCAR [55]	IJCV 2022	ResNet-50/ST50	0.873	0.611	0.745	0.488
16	GRM [6]	CVPR 2023	ViT-Large/320	0.920	0.672	0.891	0.649
17	SeqTrack [56]	CVPR 2023	ViT-Base/384-Default	0.946	0.690	0.890	0.641
18	SMAT [57]	WACV 2024	MobileViTv2/Default	0.893	0.638	0.829	0.581
19	AQATrack [58]	CVPR 2024	HiViT-Base/384-Default	0.937	0.686	0.869	0.632
20	EVPTrack [59]	AAAI 2024	HiViT-Base/224-Default	0.932	0.677	0.882	0.642
21	ARTrackV2 [60]	CVPR 2024	HiViT-Base/256-GOT-10k	0.912	0.623	0.876	0.589
22	MCITrack [43]	AAAI 2025	ViT-Base/256-GOT-10k	0.951	0.703	0.878	0.646
23	UST-Net	Ours	ViT-Base/256-Default	-	-	0.942	0.717

TABLE III

COMPARISON WITH SOTA HS TRACKERS ON HOTC DATASET. THE BEST TWO RESULTS ARE SHOWN IN RED AND BLUE FONTS

	MHT	BAE-Net	SST-Net	PHTrack	SEE-Net	SPIRIT	MMF-Net	DaSSP-Net	HotMoE	UST-Net
Venues	TIP 2020	ICIP 2020	WHISPERS 2021	TGRS 2024	TIP 2023	TGRS 2024	TGRS 2024	PR 2025	TMM 2025	Ours
AUC	0.516	0.606	0.623	0.660	0.666	0.679	0.691	0.682	0.705	0.717
DP@20	0.881	0.877	0.916	0.919	0.932	0.925	0.932	0.917	0.935	0.942

transformers (AQATrack) [58], explicit visual prompts for visual object tracking (EVPTrack) [59], prompting autoregressive tracker (ARTrackV2) [60], and exploring enhanced contextual information for video-level object tracking (MCITrack) [43]. These trackers are evaluated under two settings: on the original RGB videos and on their FAC HS counterparts. The corresponding precision and success curves are illustrated in Fig. 4, providing a more comprehensive comparison across different trackers.

Among the recent prompt- and memory-enhanced trackers in Table II, EVPTrack, ARTrackV2, and MCITrack achieve FAC success scores of 0.642, 0.589, and 0.646, respectively, compared with 0.717 for UST-Net. Other recent foundation-model and memory-enhanced trackers discussed in Section II-A are not reported in Table II because public HOTC results under a compatible HS-input protocol are unavailable, and adapting them to HS inputs would require method-specific retraining.

Overall, RGB trackers perform significantly better on RGB sequences than on FAC inputs. For example, SiamRPN++ achieves 0.866/0.621 (precision/success) on RGB videos, but its performance drops to 0.761/0.531 on FAC sequences, illustrating the limitations of transferring RGB models to HS scenarios due to spectral domain mismatch. Trackers such as SiamBAN and SiamCAR show relative robustness on FAC videos, yet their performance is still constrained by the lack of spectral-specific modeling.

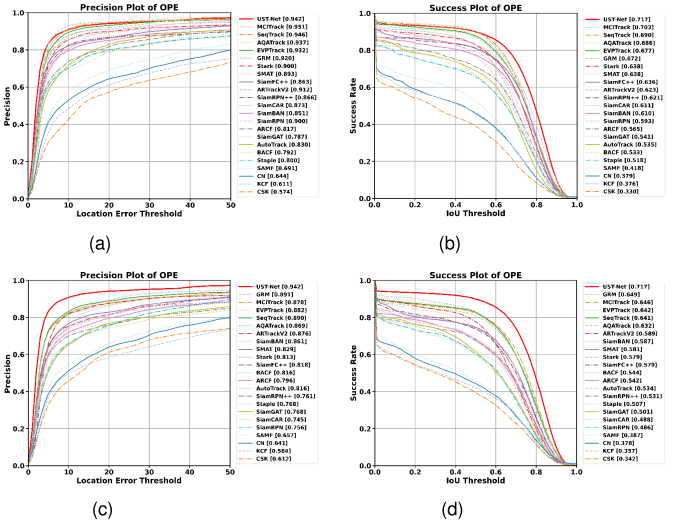


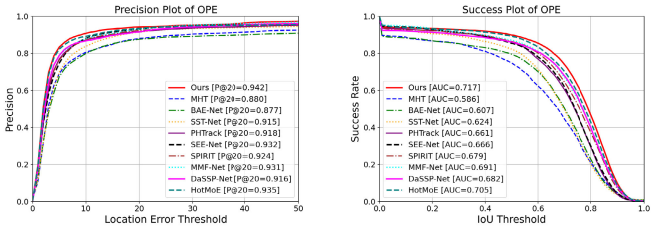
Fig. 4. Precision and success plots of the compared RGB trackers on the HOTC dataset RGB/FAC modality. (a) RGB precision. (b) RGB success. (c) FAC precision. (d) FAC success.

In contrast, our UST-Net directly leverages raw HS inputs via a cross-modal joint encoding strategy, achieving the best performance with a precision of 0.942 and a success score of 0.717. The proposed CMP module enables spatial–spectral token-level fusion, while the TSI and API modules further enhance contextual understanding across frames and feature layers.

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TABLE IV
ATTRIBUTE-BASED COMPARISON WITH SOTA HS TRACKERS ON HOTC DATASET. THE BEST TWO RESULTS ARE SHOWN IN RED AND BLUE FONTS

Attributes	MHT	BAE-Net	SST-Net	PHTrack	SEE-Net	SPIRIT	MMF-Net	DaSSP-Net	HotMoE	UST-Net
Background Clutter (BC)	0.606	0.651	0.685	0.679	0.705	0.715	0.708	0.655	0.695	0.697
Deformation (DEF)	0.664	0.680	0.700	0.674	0.742	0.723	0.724	0.765	0.710	0.778
Fast Motion (FM)	0.542	0.608	0.561	0.735	0.711	0.715	0.704	0.708	0.687	0.723
In-Plane Rotation (IPR)	0.670	0.699	0.696	0.725	0.742	0.760	0.782	0.762	0.719	0.800
Illumination Variation (IV)	0.477	0.525	0.502	0.578	0.566	0.567	0.592	0.659	0.678	0.722
Low Resolution (LR)	0.475	0.489	0.462	0.583	0.626	0.629	0.627	0.655	0.732	0.695
Motion Blur (MB)	0.560	0.593	0.536	0.730	0.702	0.723	0.764	0.724	0.730	0.756
Occlusion (OCC)	0.564	0.555	0.594	0.618	0.622	0.643	0.650	0.683	0.689	0.736
Out-of-Plane Rotation (OPR)	0.644	0.701	0.699	0.709	0.729	0.751	0.769	0.763	0.748	0.791
Out-of-View (OV)	0.626	0.517	0.480	0.604	0.633	0.689	0.602	0.815	0.600	0.867
Scale Variation (SV)	0.574	0.604	0.602	0.646	0.649	0.665	0.680	0.689	0.727	0.740



(a) (b)

Fig. 5. (a) Precision and (b) success plots of all the compared HS trackers on the HOTC dataset.

b) HS trackers comparison: We further compare UST-Net with nine SOTA HS trackers, including MHT [7], BAE-Net [13], SST-Net [39], PHTrack [15], SEE-Net [34], SPIRIT [16], MMF-Net [14], DaSSP-Net [63], and HotMoE [19], as summarized in Table III. These methods can be broadly categorized into two main groups.

Spectral Regrouping-Based Trackers: BAE-Net, PHTrack, SEE-Net, SPIRIT, and HotMoE convert HS cubes into multiple three-channel representations to utilize RGB-pretrained backbones or prompting strategies. This spectral simplification reduces computational overhead but sacrifices global spectral continuity and interband correlations, often leading to degraded performance in challenging scenes.

Fusion and Representation Learning-Based Trackers: DaSSP-Net employs a domain-adaptive self-supervised framework to extract transferable spatial–spectral representations. MMF-Net uses material-guided unmixing and multiview fusion to integrate complementary spectral views for enhanced tracking robustness.

As illustrated in Fig. 5, UST-Net consistently outperforms all other SOTA HS trackers on the HOTC dataset. In the precision plot [Fig. 5(a)], our method achieves the highest distance precision at 20 pixels (DP@20) score of 0.942, while the next best (HotMoE) reaches 0.935. Similarly, in the success plot [Fig. 5(b)], UST-Net leads with an area under the curve (AUC) of 0.717, surpassing HotMoE (0.705) and MMF-Net (0.691).

These results highlight the robustness of our framework under varying location error and IoU thresholds. The evident margin of superiority demonstrates that our model not only tracks more precisely but also maintains higher spatial alignment consistency across diverse and challenging scenarios.

c) Attribute-based analysis: As reported in Table IV, UST-Net achieves top or near-top performance across all 11 evaluated attributes. In particular.

1) *OCC:* UST-Net achieves an AUC of 0.736, a 5.3% improvement over the previous best (DaSSP-Net: 0.683). This gain reflects the model's strong temporal memory and ability to maintain target identity under partial or full occlusions, largely attributed to the TSI module.
2) *OV:* UST-Net obtains a new SOTA AUC of 0.867, outperforming the second-best (DaSSP-Net: 0.815) by 5.2%. The API and global temporal modeling enable effective recovery when the target reenters the scene after being OV.
3) For other attributes such as DEF, FM, and SV, UST-Net maintains top-tier performance, demonstrating robustness across diverse tracking challenges.

d) Visual result analysis: Fig. 6 presents qualitative comparisons on five representative HOTC sequences. UST-Net demonstrates.

1) Precise localization despite dense backgrounds and abrupt object motions.
2) Robust appearance modeling under severe blur, illumination shifts, and shape deformations.
3) Reliable temporal consistency, particularly in occlusion and OV scenarios—highlighting the contribution of the TSI and API modules.

These visual observations are consistent with the quantitative improvements under occlusion and target reappearance, indicating that coordinated spatial–spectral prompting and temporal-state interaction improve localization stability in challenging sequences.

The attribute-level results in Table IV also reveal several remaining limitations. UST-Net is less competitive under LR, BC, and FM. When a target occupies only a few pixels or shares similar spectral responses with nearby distractors, the FAC and multiband observations may become simultaneously ambiguous, limiting the discriminative prompt constructed by CMP and causing API to modulate subsequent stages from unreliable features. Abrupt displacement may also move the target beyond the local search region; TSI preserves recurrent appearance context but does not explicitly extrapolate motion or perform global redetection, allowing stale states to induce tracking drift. These cases motivate uncertainty-aware state

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Fig. 6. Qualitative results: 1) paper, attributes: BC, IPR; 2) Pedestrian2, attributes: LR, OCC, IV, DEF; 3) fruit, attributes: BC, OCC; 4) ball, attributes: MB, OCC, SV; and 5) trucker, attributes: OCC, IV, SV, OV. Bounding boxes are displayed on the FAC images. The current frame is shown in the top-left corner.

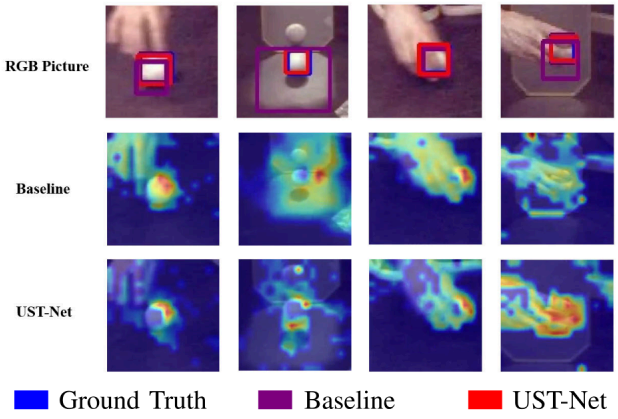


Fig. 7. Qualitative comparison of response maps on the Ball&Mirror10 sequence under different challenges: partial occlusion, OV reappearance, occlusion confusion, and full occlusion. UST-Net maintains accurate focus on the target, outperforming the baseline.

updating and motion-guided search adaptation to suppress unreliable memory and improve target recovery.

2) Results on HOT2024 Dataset: To further evaluate the generalization ability of UST-Net, we conduct experiments on the HOT2024 benchmark, which includes three spectral variants: HOT2024-VIS, HOT2024-NIR, and HOT2024-RedNIR. Table V reports the AUC and DP@20 scores for all compared methods.

1) *HOT2024-VIS:* On this VIS subset with 67 videos, UST-Net achieves the best performance, obtaining an AUC of 0.674 and DP@20 of 0.825. Compared to the second-best tracker HS mamba for HS object tracking (HyMamba) [45] (0.589/0.751), our model shows a relative improvement of 8.5% in AUC and 7.4% in precision.
2) *HOT2024-NIR:* On the NIR subset consisting of 30 videos, UST-Net achieves competitive performance with an AUC of 0.774 and DP@20 of 0.932. While HyMamba [45] slightly outperforms UST-Net in this specific domain (0.788/0.939), our model remains highly robust and consistently ranks among the top methods across different spectral variants. In particular, UST-Net significantly outperforms other baselines such as MMF-Net and SPIRIT, demonstrating its strong capability to capture subtle spatial–spectral variations in the NIR domain.
3) *HOT2024-RedNIR:* This subset poses additional challenges due to reduced spectral dimensionality. Despite this, UST-Net delivers the best results with an AUC of 0.594 and DP@20 of 0.749, outperforming the second-best HyMamba [45] (0.552/0.689) by substantial margins.

Overall, these experiments indicate that UST-Net achieves a strong and balanced performance across VIS, NIR, and RedNIR domains, showcasing its generalization ability and robustness in diverse spectral scenarios.

D. Ablation Study

To validate the effectiveness of each proposed component in UST-Net, we conduct comprehensive ablation studies on the HOTC dataset. We progressively add each module to demonstrate their individual and combined contributions to the overall tracking performance.

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TABLE V
COMPARISON WITH SOTA METHODS ON HOT2024 DATASET. THE BEST TWO RESULTS ARE SHOWN IN RED AND BLUE FONTS

Method	Venue	HOT2024-VIS		HOT2024-NIR		HOT2024-RedNIR	
		AUC	DP@20	AUC	DP@20	AUC	DP@20
SEE-Net	TIP 2023	0.396	0.560	0.509	0.769	0.383	0.521
Trans-DAT	CVPR 2021	0.397	0.524	0.587	0.753	0.423	0.547
SPIRIT	TGRS 2024	0.319	0.409	0.655	0.823	0.377	0.516
MMF-Net	TGRS 2024	0.482	0.645	0.701	0.875	0.388	0.521
PHTrack	TGRS 2024	0.303	0.410	0.526	0.731	0.262	0.364
SENSE	IF 2024	0.298	0.398	0.561	0.765	0.364	0.465
ViPT	CVPR 2023	0.540	0.716	0.692	0.852	0.493	0.650
VP-HOT	WHISPERS 2023	0.540	0.714	0.769	0.916	0.388	0.500
SeqTrack	CVPR 2023	0.519	0.666	0.620	0.796	0.385	0.508
SSTtrack [61]	IF 2025	0.386	0.482	0.712	0.888	0.511	0.659
UBSTrack [62]	TGRS 2025	0.534	0.681	0.725	0.861	0.496	0.627
HyMamba	TIP 2025	0.589	0.751	0.788	0.939	0.552	0.689
UST-Net	Ours	0.674	0.825	0.774	0.932	0.594	0.749

TABLE VI
ABLATION STUDY OF UST-NET ON HOTC DATASET

	AUC	ΔAUC	DP@20	ΔDP@20
Baseline	0.558	-	0.818	-
Baseline+CMP	0.675	0.117	0.908	0.090
Baseline+CMP+TSI	0.685	0.010	0.917	0.009
Baseline+CMP+TSI+API	0.717	0.032	0.942	0.025

The baseline model is the original OSTrack pretrained on large-scale RGB datasets and directly evaluated on the HOTC FAC sequences. For each module-enhanced configuration, only the included newly introduced modules are optimized on the HOT2024 training set following the same protocol described in Section IV-A, while the pretrained backbone and prediction head remain frozen.

Table VI presents the quantitative results of our ablation study. Adding the CMP module to the baseline yields the most significant performance improvement, with AUC increasing from 0.558 to 0.675 (+0.117) and DP@20 from 0.818 to 0.908 (+0.090). This substantial gain demonstrates the critical importance of cross-modal fusion between HS and FAC representations. The CMP module serves as the foundation of our framework by enabling effective spatial–spectral token-level interactions, which are essential for bridging the domain gap between RGB-pretrained features and HS characteristics.

The addition of TSI further improves performance to 0.685 AUC (+0.010) and 0.917 DP@20 (+0.009). Although the overall gain is modest, it verifies that recurrent temporal-state propagation provides complementary information beyond CMP. Table IV evaluates the complete model rather than TSI in isolation; nevertheless, the strong results under OCC (0.736) and OV (0.867) are consistent with the intended role of TSI in preserving historical context across frames and supporting target recovery after temporary occlusion or OV events.

The final addition of API yields notable performance gains, reaching 0.717 AUC (+0.032) and 0.942 DP@20 (+0.025). This improvement indicates that stage-conditioned residual modulation of the TSI-refined representation further benefits feature extraction at multiple backbone depths. The successive gains in Table VI suggest that CMP, TSI, and API make complementary contributions within the complete model: CMP provides spatial–spectral prompts, TSI introduces recurrent temporal context, and API uses the refined context to guide downstream feature extraction. Overall, the complete model improves AUC from 0.558 to 0.717 (+0.159) and DP@20 from 0.818 to 0.942 (+0.124) over the baseline.

Unlike feature concatenation or temporal self-attention over a predefined history window, TSI recursively updates a fixed-size stage-specific state and exchanges it with current tokens through paired injection and extraction. It therefore avoids storing an expanding set of past-frame tokens. The gain introduced by TSI in Table VI, together with the limited overall increase from 29.0 to 29.1 GFLOPs reported in Table I, supports this design choice.

API uses $M = 8$ latent prompt components, which are averaged into a single context vector before residual injection and therefore do not alter the token length of subsequent backbone stages. The Mamba and API settings are fixed whenever the corresponding modules are enabled, so Table VI assesses their module-level contributions under a consistent configuration. MCITrack is included in Table II only as a method-level reference to an alternative Mamba-based memory design, since differences in input adaptation and overall architecture preclude a controlled comparison with TSI.

1) Feature Map Analysis in a Challenging Scenario: We conduct a qualitative feature map analysis on the most challenging sequence Ball&Mirror10 from the HOT2024 benchmark. This scenario includes several challenging factors, such as occlusion, OV cases, reflections, and small target size. As shown in Fig. 7, we compare the response maps of UST-Net and the baseline method under various conditions.

1) *Column 1–Partial Occlusion:* When the target is partially occluded by a hand, UST-Net maintains a focused response on the target region, while the baseline generates a more dispersed activation map with weaker robustness to occlusion.
2) *Column 2–OV and Reentry:* After the target temporarily disappears and reenters the field of view, UST-Net rapidly relocates the target by leveraging global temporal modeling. The baseline fails

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to recover the target and produces scattered, noisy responses.

3) *Column 3–Occlusion Confusion:* When both the hand and the ball are present, the baseline tends to focus on the interfering hand, whereas UST-Net accurately attends to the actual target, showing stronger discriminative ability.
4) *Column 4–Full Occlusion:* Even under complete occlusion, UST-Net preserves meaningful activation near the expected target location due to its use of long-term temporal information. In contrast, the baseline fails to generate any informative response.

These visualizations demonstrate that UST-Net effectively maintains target focus under extreme visual challenges, including occlusion, OV cases, and distracting interference, which highlights the benefit of its spatio-temporal design.

V. Conclusion

This work presented UST-Net, a unified spatial–spectral–temporal framework for HS object tracking. UST-Net establishes a progressive information pathway in which CMP organizes co-registered FAC and grouped multiband observations into fixed-format, spectrum-preserving prompts; TSI couples the prompt-enhanced token stream with cross-frame Mamba hidden states and cross-stage semantic context; and API converts the temporally refined representations into stage-conditioned residual modulation for subsequent feature extraction. Through this pathway, spectral evidence, historical target states, and hierarchical visual representations are coordinated within a shared transformer tracker. Experiments on the HOTC and HOT2024 datasets demonstrate state-of-the-art performance across the evaluated spectral configurations, with particularly strong results under occlusion and OV conditions, supporting the effectiveness of the proposed information pathway.

Future work will investigate uncertainty-aware state updating and motion-guided search adaptation to suppress unreliable temporal context and improve target recovery under LR, BC, and abrupt motion. More resource-efficient implementations will also be explored to facilitate deployment across diverse HS platforms.

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